

# CDS 101 Disney Project

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2026-05-11

```
### Shows Each Column with examples
```

```
glimpse(disney)
```

```
## Rows: 1,450
## Columns: 12
## $ show_id      <chr> "s1", "s2", "s3", "s4", "s5", "s6", "s7", "s8", "s9", "s1~
## $ type         <chr> "Movie", "Movie", "Movie", "Movie", "TV Show", "Movie", "~
## $ title        <chr> "Duck the Halls: A Mickey Mouse Christmas Special", "Erne~
## $ director     <chr> "Alonso Ramirez Ramos, Dave Wasson", "John Cherry", "Kare~
## $ cast         <chr> "Chris Diamantopoulos, Tony Anselmo, Tress MacNeille, Bil~
## $ country      <chr> "", "", "United States", "", "", "United States", "", "Un~
## $ date_added   <chr> "November 26, 2021", "November 26, 2021", "November 26, 2~
## $ release_year <int> 2016, 1988, 2011, 2021, 2021, 2021, 2021, 2015, 2019, 200~
## $ rating       <chr> "TV-G", "PG", "TV-G", "TV-PG", "", "PG-13", "TV-14", "TV~~
## $ duration     <chr> "23 min", "91 min", "23 min", "41 min", "1 Season", "94 m~
## $ listed_in    <chr> "Animation, Family", "Comedy", "Animation, Comedy, Family~
## $ description  <chr> "Join Mickey and the gang as they duck the halls!", "Sant~
```

```
### Shows each Variable
```

```
summary(disney)
```

```
##      show_id          type          title          director
## Length:1450      Length:1450      Length:1450      Length:1450
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##      cast          country          date_added          release_year
## Length:1450      Length:1450      Length:1450      Min.   :1928
## Class :character Class :character Class :character      1st Qu.:1999
## Mode  :character Mode  :character Mode  :character      Median :2011
##                                     Mean   :2003
##                                     3rd Qu.:2018
```

```
##
##      rating      duration      listed_in      Max.      :2021
##      Length:1450      Length:1450      Length:1450      Length:1450
##      Class :character      Class :character      Class :character      Class :character
##      Mode  :character      Mode  :character      Mode  :character      Mode  :character
##
##
##
```

2.)

```
### Converts the release year variable from character to a numeric value
disney <- disney %>%
  mutate(release_year = as.numeric(release_year))
```

```
### Converts the duration variable from character to a numeric value
disney <- disney %>%
  mutate(duration_num = parse_number(duration))
```

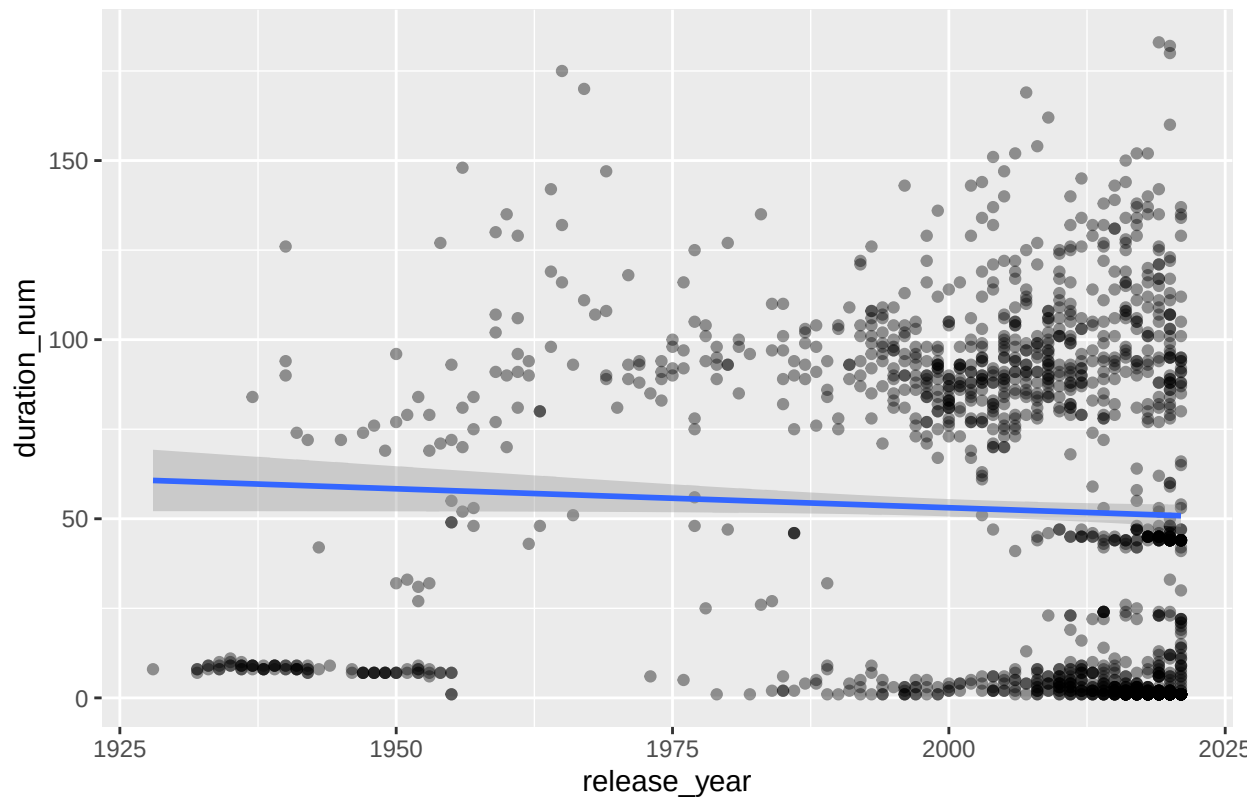
```
### Transforms the ratings and listed_in variables into factors which makes them categorical
disney <- disney %>%
  mutate(
    rating = as.factor(rating),
    listed_in = as.factor(listed_in)
  )
```

3.) Exploring Relationships:

```
#Creates a scatter plot looking at the relationship between the release year
#and the duration of the movie to see if there is any correlation between the two.
ggplot(disney, aes(x = release_year, y = duration_num)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm") +
  labs(title = "Duration vs Year of Release")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Duration vs Year of Release



ii.)

*# Adds a new column named period which splits release years into different periods  
# of time being 25 years apart from each other.*

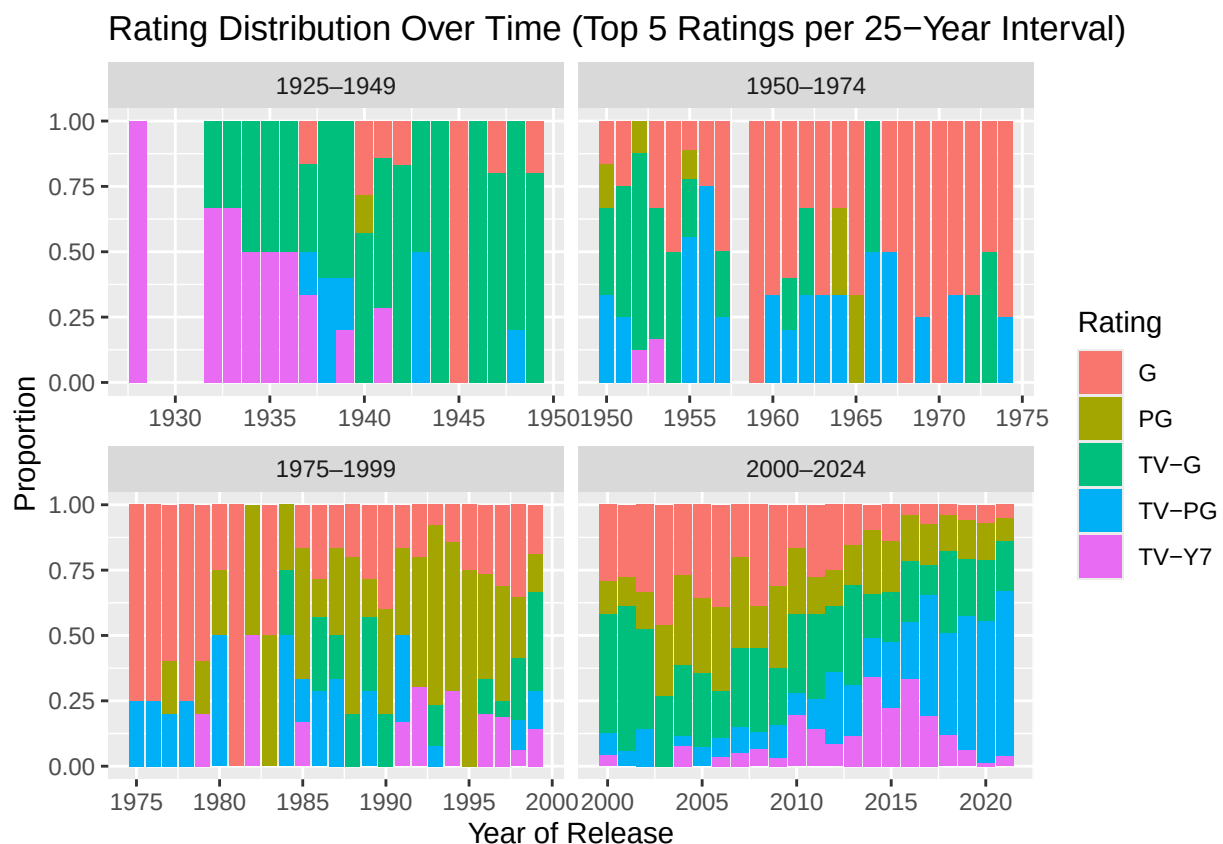
```
disney <- disney %>%
  mutate(period = cut(
    release_year,
    breaks = c(1900, 1925, 1950, 1975, 2000, 2025),
    labels = c("1900-1924", "1925-1949", "1950-1974", "1975-1999", "2000-2024"),
    right = FALSE
  ))
```

*# Creates a new data set looking at the amount of movies spread based on  
# two different factors.  
# The factors being time period and rating of the movie.  
# For example, This just gives the total amount of movies for rated G during  
# the years of 2000-2024*

```
top_ratings <- disney %>%
  count(period, rating, sort = TRUE) %>%
  group_by(period) %>%
  slice_max(n, n = 5) %>%
  ungroup()
```

```
#New dataset making sure all the values match from top_ratings using the semi_join
#function
ratings_top <- disney %>%
  semi_join(top_ratings, by = c("period", "rating"))
```

```
#Creates a bar plot looking at the distribution of age rating throughout the years.
ratings_top %>%
  count(period, release_year, rating) %>%
  ggplot(aes(x = release_year, y = n, fill = rating)) +
  geom_col(position = "fill") +
  facet_wrap(~ period, scales = "free_x") +
  labs(
    title = "Rating Distribution Over Time (Top 5 Ratings per 25-Year Interval)",
    x = "Year of Release",
    y = "Proportion",
    fill = "Rating"
  )
)
```



iii.)

```
# Creates a new data set with new rows based on the listed_in Variable.
# This new data set creates new rows based on the different genres of each movie.
```

```

# For example the first movie in the data set
# "Duck the Halls: A Mickey Mouse Christmas Special"
# is split into two different two rows with one having animation and the other being family.
# Not all rows will be split, but all rows will keep the other same data
# such as rating and release year.
disney_genre <- disney %>%
  separate_rows(listed_in, sep = ", ") %>%
  mutate(listed_in = as.factor(listed_in))

```

```

# This does the same thing as earlier, adding a new column named period.
# This adds a column based on the release date of the movie,
# being 25 years apart from each other.
disney_genre <- disney_genre %>%
  mutate(period = cut(
    release_year,
    breaks = c(1900, 1925, 1950, 1975, 2000, 2025),
    labels = c("1900-1924", "1925-1949", "1950-1974", "1975-1999", "2000-2024"),
    right = FALSE
  ))

```

```

# Does the same thing earlier and counts the total amount of movies
# based on rating sorted year
top_genres <- disney_genre %>%
  count(period, listed_in, sort = TRUE) %>%
  group_by(period) %>%
  slice_max(n, n = 5) %>%
  ungroup()

```

```

#New dataset making sure all the values match from disney_genre using the semi_join
#function
disney_top <- disney_genre %>%
  semi_join(top_genres, by = c("period", "listed_in"))

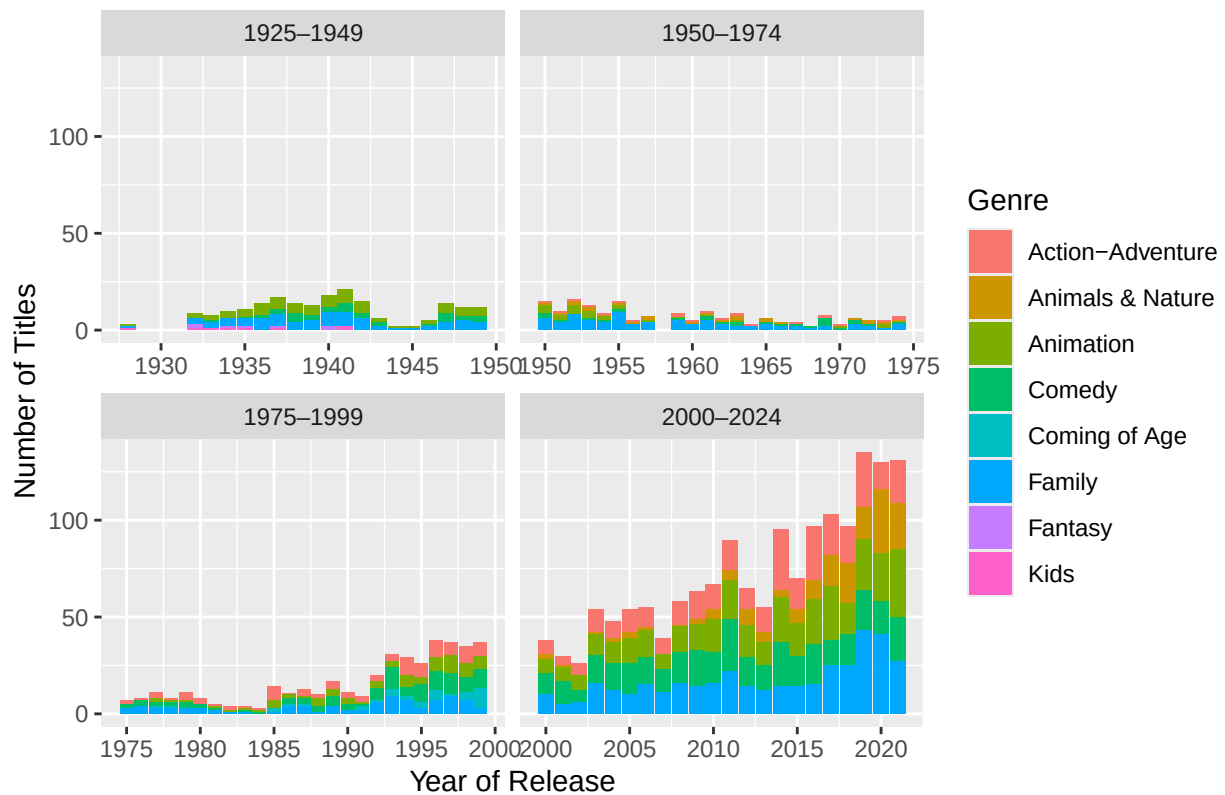
```

```

# Creates multiple bar plots looking at the top 5 genres
# at a with a range of 25 years.
# Shows the relationship between genres and intervalss of release.
disney_top %>%
  count(period, release_year, listed_in) %>%
  ggplot(aes(x = release_year, y = n, fill = listed_in)) +
  geom_col() +
  facet_wrap(~ period, scales = "free_x") +
  labs(
    title = "Top 5 Genres per 25-Year Interval",
    x = "Year of Release",
    y = "Number of Titles",
    fill = "Genre"
  )

```

Top 5 Genres per 25-Year Interval



iv.)

*#Separates the genres in the table into one individual rating, instead of two or more.*

```
ratings_top <- disney %>%
  separate_rows(listed_in, sep = ", ") %>%
  mutate(listed_in = as.factor(listed_in))
```

*#Identifies the top 5 genres*

```
ratings_top <- disney_genre %>%
  count(listed_in, sort = TRUE) %>%
  slice_max(n, n = 5) %>%
  pull(listed_in)
```

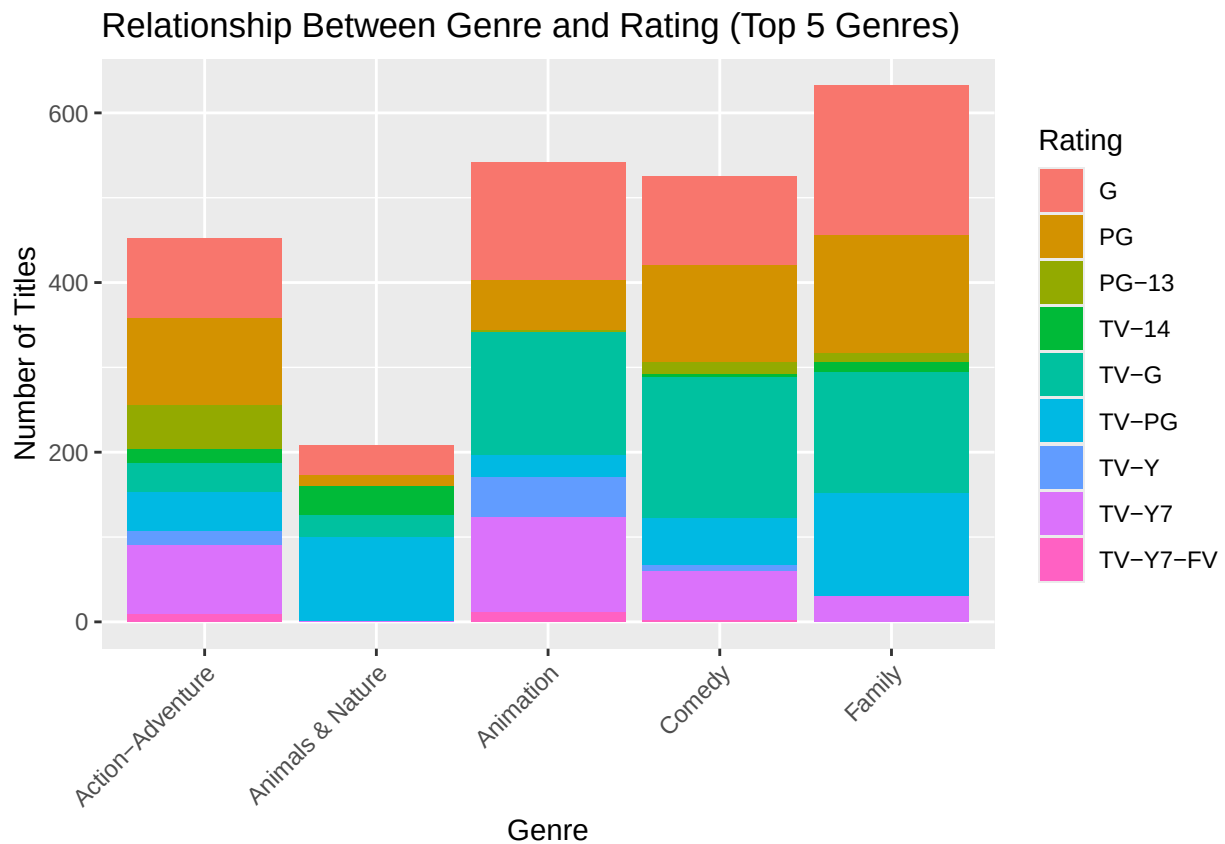
*#Filters the data set to only the top 5 genres.*

```
ratings_top <- disney_genre %>%
  filter(listed_in %in% ratings_top)
```

*#Makes bar chart to see relationship between the genre and rating.*

```
ratings_top %>%
  count(listed_in, rating) %>%
  ggplot(aes(x = listed_in, y = n, fill = rating)) +
  geom_col() +
```

```
labs(
  title = "Relationship Between Genre and Rating (Top 5 Genres)",
  x = "Genre",
  y = "Number of Titles",
  fill = "Rating"
) +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



4.)

```
# fits the duration and release year into a linear regression
# model and provides statistics such as p values, estimates,
# r.squared, and sigma
model_duration <- lm(duration_num ~ release_year, data = disney)
tidy(model_duration)
glance(model_duration)
```

```
# Fits the rating year into a linear regression model
# based on the release years and provides the same statistics
model_rating <- lm(as.numeric(rating) ~ release_year, data = disney)
tidy(model_rating)
```

```
# Fits the rating into a linear regression model
# based on the genre and provides the same statistics
model_rating_vs_genre <- lm(as.numeric(rating) ~ listed_in, data = ratings_top)
tidy(model_rating_vs_genre)
```

5.)

```
# Does the same thing as the other code chunks, basically asking
# How release year, rating, and genre impact the movie duration
model_multi <- lm(duration_num ~ release_year + rating + listed_in, data = disney)
tidy(model_multi)
glance(model_multi)
```

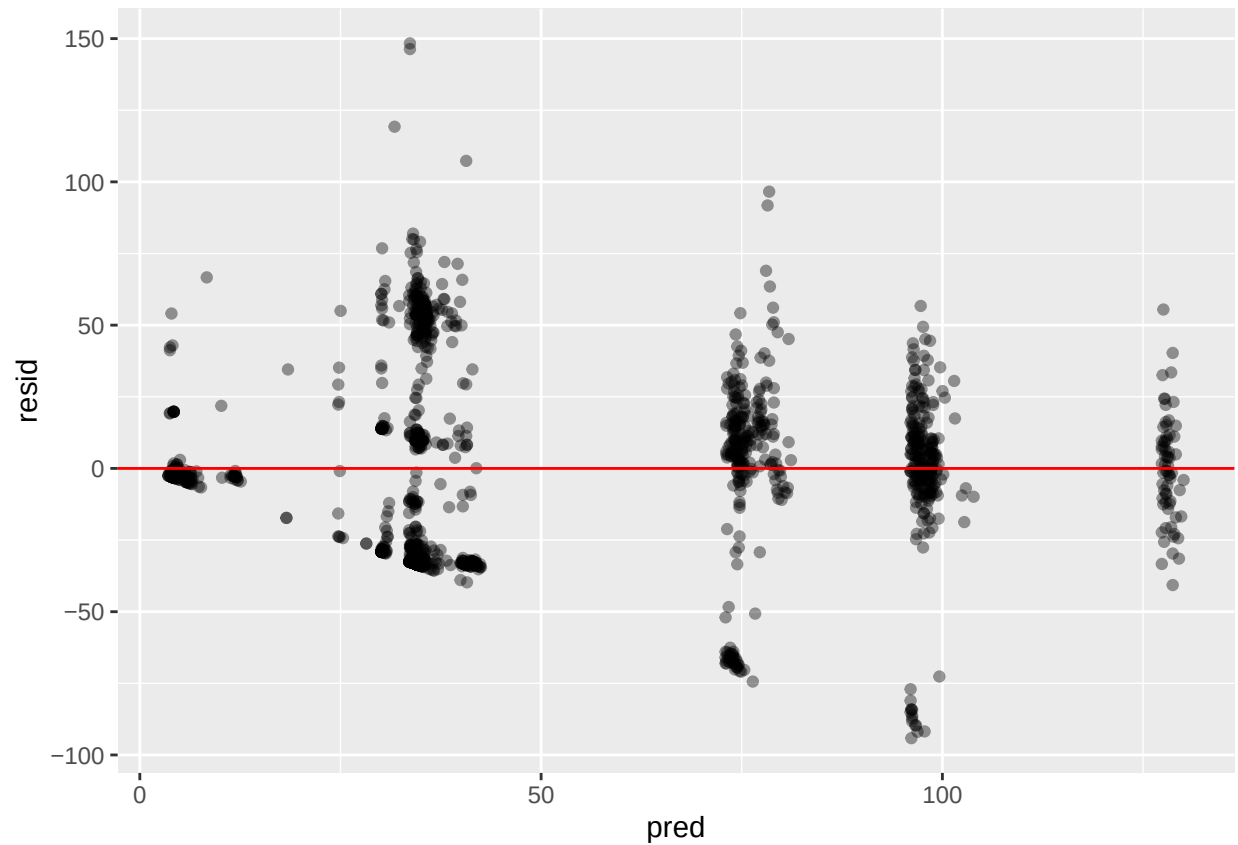
```
# This creates a linear regression model looking and how
# only release year and rating predict duration.
# and provides statistics like previous code chunks
model_multi1 <- lm(duration_num ~ release_year + rating, data = disney)
tidy(model_multi)
glance(model_multi)
```

6.)

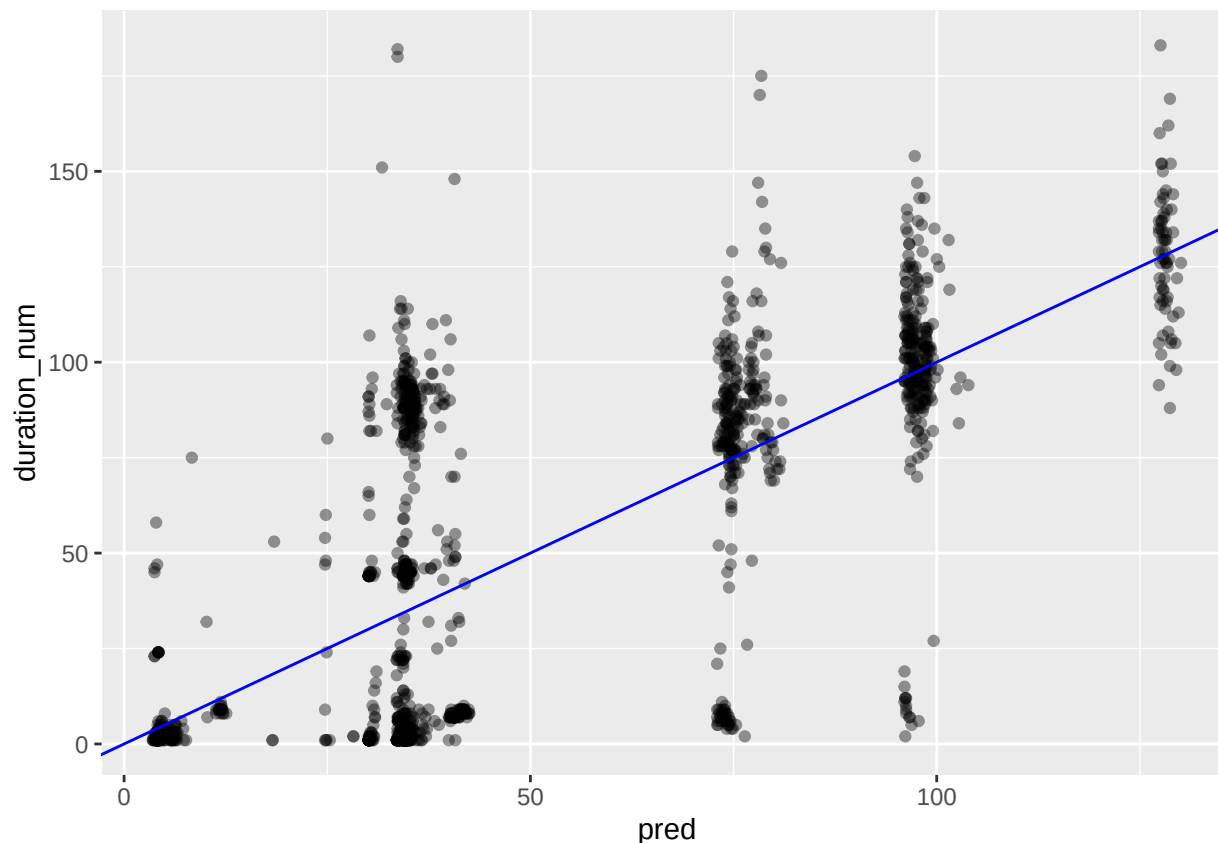
```
# Adds the pred and resid variables from the previous
# code chunks to a new dataset named disney_pred
disney_pred <- disney %>%
  add_predictions(model_multi1, var = "pred") %>%
  add_residuals(model_multi1, var = "resid")
```

7.)

```
# Plots residual vs predicted durations
# x= predicted duration
# y = residuals (actual - predicted)
ggplot(disney_pred, aes(x = pred, y = resid)) +
  geom_point(alpha = 0.4) +
  geom_hline(yintercept = 0, color = "red")
```



```
# Plots predicted vs actual duration  
# X is the predicted duration  
# y is the acutal duration of the movie  
ggplot(disney_pred, aes(x = pred, y = duration_num)) +  
  geom_point(alpha = 0.4) +  
  geom_abline(color = "blue")
```



## #Conclusion

This analysis provides a clear understanding of how Disney+ content has evolved across time, genre, and rating, and how these factors relate to duration. By breaking the data-set into 25-year intervals and focusing on the most common genres and ratings within each period, the visualizations revealed to us meaningful historical shifts in content production. The genre vs. rating comparison further showed that certain genres consistently align/have a relationship with specific rating categories, while others show a wider spread, indicating to us that they have more variability in how they are classified.

The multiple regression model reinforced these findings by showing that genre and rating are strong predictors of duration, explaining over 80% of the variation in run-time. The model performed well, and the residual analysis showed no major violations of assumptions, supporting its validity. Taken together, the graphics and statistical modeling show that genre and rating play central roles in shaping Disney+ content and are key to understanding differences in duration across titles.

Our github repository link is listed below: <https://github.com/Sommy121/CDS-Project-AJS->